

Art + Code: Case Study in Expanding Access with p5.js

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Figure 1: p5.js sketch still by Ivan Ortega, General Art Teacher at Pebble Hills High School, El Paso, TX during Art + Code.

Abstract

p5.js is a JavaScript library that provides a consistent graphics API across several extensible renderers. Widely used for making and teaching algorithmic art, it prioritizes access and provides flexibility and expressiveness. The goal of “beginner-friendliness” in p5.js includes not only supporting learners at their earliest stages, but also educators teaching in disciplines outside of computer graphics to be able to incorporate algorithmic art into their teaching. This presentation reflects on a professional development program for teachers and is delivered by a teacher who has participated in this program, and focuses on the perspective of the student impact

that integrating p5.js materials into his teaching has had on his students. The projects in this program ground the fundamentals of web-based interactive graphics in the contexts of students, to expand the imagination of what is possible with the resources they already have, and suggest new future professional possibilities.

CCS Concepts

• **Social and professional topics** → **K-12 education**; Computing literacy; *Computing education programs*.

Keywords

Education, Fundamentals, Web-based Interaction, JavaScript

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Table 1: Metadata for the “A Walk in the Neighborhood” [Art + Code 2025b] assignment, which teachers complete as part of Art+Code and can then incorporate into their teaching. Training includes both 6-8 grade variant, and 9-12.

Summary	A Walk in the Neighborhood
Learning Outcomes	Understand files and variables, apply canvas grid, keyboard events, and animation/draw loop to create an interactive digital collage
Classification(s)	Fundamentals, Image Manipulation
Audience	6th - 8th grade: loading and displaying image within coordinate grid, positioning it using conditionals and mouse interaction.
Dependencies	Students should be familiar with variables, and mouse and keyboard interaction.
Prerequisites	Drawing with shape primitives using the p5.js API; p5.js sketch setup and draw loop; and JavaScript variables, introduced in two prior projects.
Strengths	The mixing of analog arts materials with digital interaction.
Weaknesses	There are many steps to take on, including preprocessing images. Sketch can take longer to load, depending on number and size of images.
Variants	The choice of image assets is the main technical variation.
Assessment	Balances artistic, conceptual, and coding skill, including application of technical terms and p5.js API elements, debugging, and documentation [Art + Code 2025a].

1 Introduction

p5.js is a JavaScript library that provides a consistent graphics API across several extensible renderers. From its inception by Lauren Lee McCarthy in 2013, it was created as an entry point for women and people of color to open source development in a field that lacked representation [Tusman et al. 2021]. In 2019, alongside the release of p5.js 1.0, the p5.js contributor community adopted the philosophy to only consider features that increase inclusion and access across multiple dimensions, including beginner-friendliness and a focus on historically marginalized communities [p5.js contributors 2019]. Currently, over 800 open source contributors have made both the library and community-maintained documentation and add-on libraries to be used in combination with p5.js, and with each other: for example, p5.spatial builds on p5.js, p5.xr, and P5Live to enable “real-time prototyping of immersive multi-user sketches directly on XR headsets using a keyboard and mouse” [Udvari 2025]. As a graphics tool, p5.js offers “a unique level of variety and control” to artists [Pagurek van Mossel and Kuksenok 2025], and is used widely in middle school, high school, and university curricula.

This presentation reflects on Art+Code [5], a professional development program for teachers in various subjects to become introduced to graphics programming through p5.js over the course

of one intensive week, and subsequent check-in calls in the newly established professional community of practice. Over the past two years, this program has supported 54 middle and high school teachers to learn to teach with p5.js. The teachers were mostly unfamiliar with programming and taught subjects outside of computing (typically, Art). The curricular materials from Art+Code were then used to bring free, web-based graphics education to communities that would not have other access to these kinds of projects.

This paper is presented by a teacher of Computer Science new to graphics who has participated in Art + Code; and focuses on the perspective of the student impact that integrating p5.js materials into his teaching has had on his students.

2 Broadening Engagement

Compared to other computing education materials, graphics projects are particularly well-suited to support a student’s sense that their work has “come alive.” The strongest signal of this is when students show their projects to their families or within their communities. Because p5.js is often the first coding environment a student encounters, it is designed to need no prior technical knowledge, and to integrate with the existing interests, practices, and context, rather than encountering computational and graphics context in isolation.

For example, “A Walk in the Neighborhood” [Art + Code 2025b] is one example project for 6th - 8th grade (Table 1). It is included in the Art + Code training, and the representative image (Fig. 1) reflects the work of one of the teachers who participated. This assignment introduces key concepts to algorithmic composition, and opens the possibility to go deeper into image manipulation using p5.js (e.g., including blending, layering, and filters as in [Pagurek van Mossel and Kuksenok 2025]) by demonstrating how to load and position assets on a 2D grid. This assignment demonstrates using materials that already have an established connection and meaning of their own: the personal artworks of students.

The starting point of p5.js does not assume familiarity or motivation in computer graphics specifically. Thus, one measure of success becomes engagement with computing, and motivation study of graphics in contexts that do not already include computer graphics education.

2.1 Removing Barriers

Barriers to access can take many forms, including conceptual and at the level of API design, in institutional and IT constraints, and varied needs of different students. The Art + Code program includes check-ins across the professional community of practice of participating teachers.

In terms of API design, the Art + Code program also connects participating teachers with maintainers of the p5.js open source project through informal feedback, among other forms of contribution and participation. For example, the entry point for asset loading uses the JavaScript `await` keyword to enforce sequential, rather than asynchronous execution. As a result, the weaknesses of this exercise include longer load times (Table 1). However, if this is reported as a barrier for learners, it can be improved by the community through more accessible integration or documentation of patterns that can be used (such as `Promise.all`). The design of the

p5.js API has followed through extensive input based on teaching practice, to allow scaffolding art and coding concepts.

Aside from API design, access to art and graphics software includes not only cost but also institutional integration in terms of IT requirements, computing capacity and wifi availability. Although p5.js is free and open source, additional work may be needed to make it available. Nevertheless, JavaScript web-based creative code remains accessible on many platform that might be available in a school.

3 Discussion

In the example assignment (Figure 1, Table 1), the degrees of freedom remain intentionally close to artistic and personal expression, and incorporate interaction as early as possible. These allow student engagement in contexts beyond dedicated computer graphics teaching. The choice of image assets is the main technical variation. Teachers of various subject matter were interested in making versions where a piece of food explored the digestive system (biology), or an explorer visited different lands (social studies).

An assignment like this serves to introduce the fundamentals of web graphics and algorithmic art as something that is in the students' space of capability, which is an important dimension of access. The aim is to expand the imagination of what is possible to do with the computing resources they already have, and open

new potential professional possibilities in the future. p5.js aims to be beginner friendly not only in making a JavaScript graphics API accessible to learners at their earliest stages, but also to educators teaching in disciplines outside of computer graphics to be able to incorporate graphics and algorithmic art into their teaching.

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